

SIBU, Malaysia

WMO: 964210

Lat: 2.262N

Lon: 111.985E

Elev: 37

StdP: 100.88

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 22.0	(c) 22.5	(d) 20.7	(e) 15.4	(f) 26.8	(g) 21.0	(h) 15.7	(i) 26.1	(j) 6.3	(k) 27.4	(l) 5.5	(m) 27.8	(n) 1.5	(o) 140	(p) 0.287

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	8.5	34.3	26.1	34.0	26.1	33.3	26.0	27.5	31.7	27.1	31.4	26.9	31.1	2.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	21.8	29.3	26.1	21.6	29.2	25.9	21.3	28.9	87.5	31.9	85.9	31.8	84.7	31.5	33.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 5.2	(b) 4.4	(c) 3.8	(e) 21.0	(f) 35.6	(g) 0.9	(h) 0.6	(i) 20.4	(j) 36.1	(k) 19.9	(l) 36.4	(m) 19.4	(n) 36.8	(o) 18.8	(p) 37.2		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			20.4	29.0	0.8	1.4	19.8	30.0	19.3	30.8	18.8	31.6	18.3	32.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.6	26.9	27.1	27.7	28.0	28.2	28.0	27.9	27.9	27.7	27.5	27.5	27.3
	DBStd	1.11	1.06	1.03	1.00	0.99	1.08	1.14	1.15	1.19	1.10	1.01	0.82	0.91
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6437	524	478	548	541	565	539	556	554	531	542	524	535
	CDD18.3	3395	266	245	289	291	307	289	298	296	281	284	274	277
	CDH23.3	29219	2007	1933	2463	2587	2887	2692	2760	2682	2437	2333	2223	2216
	CDH26.7	10875	608	624	905	983	1140	1067	1148	1107	945	847	767	734
Wind	WSAvg	1.5	1.6	1.6	1.6	1.5	1.5	1.5	1.5	1.6	1.5	1.5	1.5	1.5
Precipitation	PrecAvg	3577	481	350	349	263	232	201	173	223	216	281	357	451
	PrecMax	4947	887	675	648	432	468	372	398	445	498	530	694	718
	PrecMin	2601	187	80	167	115	132	39	20	51	76	103	158	204
	PrecStd	583	186	137	119	82	82	93	95	99	94	99	134	119
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.1	33.2	34.1	34.2	34.9	34.9	35.0	35.1	34.2	34.1	33.3	33.2
		MCWB	26.0	25.8	26.0	26.6	26.5	26.2	26.0	25.6	26.3	25.8	26.1	26.1
	2%	DB	32.0	32.2	33.1	33.6	34.0	33.9	34.0	34.1	33.4	33.0	32.5	32.2
		MCWB	25.9	25.7	26.0	26.5	26.5	26.1	25.8	25.7	26.1	25.9	26.0	25.9
	5%	DB	31.1	31.3	32.2	32.8	33.1	33.0	33.1	33.1	32.8	32.1	31.9	31.5
		MCWB	25.9	25.7	25.8	26.3	26.5	26.1	25.7	25.7	25.9	25.8	25.9	25.9
	10%	DB	30.0	30.4	31.2	31.8	32.1	32.0	32.1	32.1	31.8	31.1	30.9	30.7
		MCWB	25.6	25.5	25.8	26.2	26.4	26.0	25.6	25.7	25.9	25.9	25.9	25.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.1	27.1	27.2	27.9	27.8	27.6	27.3	27.2	27.7	27.2	27.2	27.2
		MCDB	30.7	30.8	31.3	32.2	32.0	31.8	31.9	31.9	31.8	31.2	31.0	31.2
	2%	WB	26.6	26.5	26.7	27.2	27.3	27.0	26.7	26.7	26.9	26.7	26.6	26.6
		MCDB	30.1	30.3	30.8	31.4	31.6	31.5	31.4	31.4	31.2	30.7	30.6	30.5
	5%	WB	26.2	26.1	26.4	26.7	26.9	26.6	26.4	26.3	26.4	26.2	26.3	26.2
		MCDB	29.6	29.7	30.3	30.8	31.3	30.9	30.9	30.9	30.7	30.3	30.1	29.9
	10%	WB	25.8	25.8	26.0	26.4	26.6	26.2	26.0	26.0	26.1	25.9	26.0	25.9
		MCDB	29.0	29.1	29.8	30.2	30.6	30.4	30.5	30.4	30.3	30.0	29.8	29.4
Mean Daily Temperature Range	5% DB	MDBR	7.1	7.4	8.3	8.4	8.5	8.4	8.9	8.8	8.5	8.1	8.0	7.6
		MCDBR	8.4	8.7	9.4	9.6	9.7	9.8	10.1	10.3	9.8	9.4	9.0	8.7
		MCWBR	3.3	3.2	3.3	3.5	3.5	3.4	3.4	3.5	3.7	3.6	3.5	3.3
	5% WB	MCDBR	7.6	7.8	8.6	8.8	8.9	8.9	9.3	9.3	9.0	8.5	8.3	8.0
		MCWBR	3.3	3.2	3.4	3.6	3.6	3.6	3.7	3.7	3.9	3.6	3.5	3.4
Clear-Sky Solar Irradiance	taub	0.412	0.424	0.429	0.411	0.413	0.413	0.423	0.502	0.518	0.511	0.418	0.404	0.404
	taud	2.466	2.409	2.392	2.440	2.444	2.443	2.407	2.170	2.129	2.149	2.439	2.488	2.488
	Ebn at Noon	914	907	897	894	871	859	855	805	809	819	901	916	916
	Edh at Noon	115	124	126	117	113	111	116	152	161	159	117	111	111
All-Sky Solar Radiation	RadAvg	4.31	4.64	5.04	5.12	4.92	4.83	4.94	4.96	4.67	4.56	4.58	4.46	4.46
	RadStd	0.36	0.26	0.23	0.21	0.22	0.25	0.24	0.27	0.37	0.42	0.19	0.23	0.23

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	+0.28	+0.34	+0.25	N/A	N/A	+81	+81
(47) Regional (2 neighbors)	N/A	N/A	N/A	+0.40	+0.40	+0.27	N/A	N/A	+94	+89

Nomenclature: See separate page